

PRAGYA MOHAPATRA

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SUMMARY

I am an aspiring computational astrophysicist also interested in data science application in fields like bioinformatics. I consider myself to be an efficient worker with creative thinking and leadership skills.

WORK EXPERIENCE

1. **Reading projects** under **Dr. Saptarshi Bej** in **IISER TVM** and assignments August 2023- Present
on machine learning. Took a biological dataset on healthcare and performed different types of clustering operations on it and created unsupervised learning algorithm and models of the data.

2. Summer internship on **High performance computing for astrophysical fluid dynamics** under **Dr. Vishwanath Shukla** in **IIT Kharagpur**. Performed simulations on statistical equations. Main focus on creating such simulations which can run on a GPU. May- July 2024
Project report: [Computational fluid dynamics](#)

3. Summer internship on **Analysing polarisation properties of Fast radio bursts** under **Dr. Esha Kundu** in **IIT Dhanbad**. In this project we take different fast radio bursts and observe their different properties like polarisation and their relationship with factors such as Rotation and dispersive measure. May-July 2024
Presentation ppt: [Final Data](#)

4. Project on simulating different trajectory problems of functions. Conducted extensive simulations of projectile trajectories utilizing three distinct trebuchet systems to analyze performance and efficiency under guidance of **Dr. Soumen Basak**. Developed a comprehensive trajectory model, incorporating the principles of dynamics and mechanics, and derived the corresponding Lagrangian for each system. [Report](#) June 2024-Present

EDUCATION

St. Agnes Higher Secondary School 2007-2020

- 96 % in Indian Certificate of Secondary Education.
- Subjects: English, Hindi, Bengali, GK, Science, Social Science, Moral science, Geography.

Kendriya Vidyalaya

2020-2022

- 91 % in Central Board of Secondary Education
- Science stream

IISER Thiruvananthapuram

Nov 2022- 2027

- Integrated BSc and Msc
- 8 CGPA till 4th semester.

ADDITIONAL INFORMATION

- Fluent in **English,Hindi,Odia, Bengali** and learning **German, Spanish, French**.
- **Technical skills:** proficient in programming languages like C,C++,CUDA,Python,FORTRAN,JAVA, BASIC,MATLAB, Julia, R, and MS Office Suite.
- **Writing skills:**Written multiple summaries of research papers and articles. [gamma ray bursts](#) , [articles](#)

ADDITIONAL COURSES

Ongoing:

- 1.Artificial Intelligence with Python(MIT online)
3. Nptel: Applied Computational fluid dynamics,Ordinary Differential equations, Machine learning,Astrophysical fluids.

Completed:

- 3.NPTEL :Computational Physics.
4. Upcoming:Computational Fluid dynamics(NPTEL)

OTHER PROJECTS

- Demonstration of **Casimir effect** and its explanation using quantum field theory in iiser science fest.
- Machine Learning** basic programming for data analysis in biomedical research.
- Summarising research papers on GRBS,FRBS,Accretion disks.
- [Article](#) on **accretion disks and jets**.
- Project on hypothesis testing of calculating **FID** of butterflies and drosophila attraction to banana under Prof.Hema Somanatahan and Ullasa Kodandaramaiah.

LAB EXPERIENCE

- **Basic lab techniques:**used instruments like spectrometer,meter bridge, potentiometer, magnetic bridge,in experiments on optics,mechanics,electrodynamics and thermodynamics.
 - **Computational :** PASCO CAPSTONE, LaTeX, Anaconda, Ubuntu ,Linux ,UNIX functions,IDLE,Spyder,VSL Studio and Text editor .
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- **SOFTWARES:** FAST,GADGET-2,XSPEC,CIGALE

EXTRACURRICULAR

- Volunteering in Physics Club and IGEM in IISER Tvm
 - Anchoring in various events in college and school.
 - Playing sports like volleyball in intra college matches.
 - Taught school and 1st year college students computational skills and basic coding skills.
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REFERENCES:

1.Dr.Saptarshi Bej, Assistant Professor School of Data Science,IISER TVM

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2. Dr. Vishwanath Shukla, Department of Physics, IIT Kharagpur.

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3. Dr. Esha Kundu , Department of physics, IIT Dhanbad

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